Co-Director: Agentic Generative Video Storytelling

Yale Song, Yiwen Song, Nick Losier, Nathan Hodson, Ye Jin, Rhyard Zhu et al. | Google

https://arxiv.org/abs/2604.24842

Research Questions

- How can we formalize video storytelling as a global optimization problem to prevent semantic drift and cascading failures in multi-agent generative pipelines?
- How can hierarchical parameterization using multi-armed bandits balance exploration of novel narrative strategies with exploitation of effective creative configurations?
- How can we rigorously evaluate end-to-end video storytelling capabilities under strict constraints like brand fidelity and demographic targeting?

Core Idea

Current video generation systems suffer from semantic drift and cascading failures when chaining multiple generative modules - early errors propagate downstream and break narrative coherence. Co-Director treats video storytelling as a global optimization problem using a multi-armed bandit to sample high-level creative directions (strategy, narrative mode, aesthetic style) that are then propagated to all sub-agents. This ensures all modules work toward a unified creative vision rather than making independent decisions. The key insight is hierarchical parameterization: instead of optimizing individual prompts, optimize global creative parameters that steer the entire pipeline, while using local self-refinement loops to fix immediate issues.

Key Formula

Keyframe Generation

$$K = \{k_0, k_1, \ldots, k_{N-1}\} \text{ where } k_i = \Phi_{frame}(s_i, V, \theta)$$

K is the set of keyframes, k_i is the keyframe for scene i, s_i are scene descriptors, V are visual assets, θ is the creative configuration sampled by MAB. This shows how global creative parameters θ condition local keyframe generation.

Video Clip Generation

$$C = \{c_0, c_1, \ldots, c_{N-1}\} \text{ where } c_i = \Phi_{video}(s_i, k_i, \theta)$$

C is the set of video clips, c_i is the clip for scene i conditioned on scene descriptors s_i, keyframe k_i, and global creative configuration θ. Demonstrates hierarchical conditioning from global parameters to local generation.

Key Figure

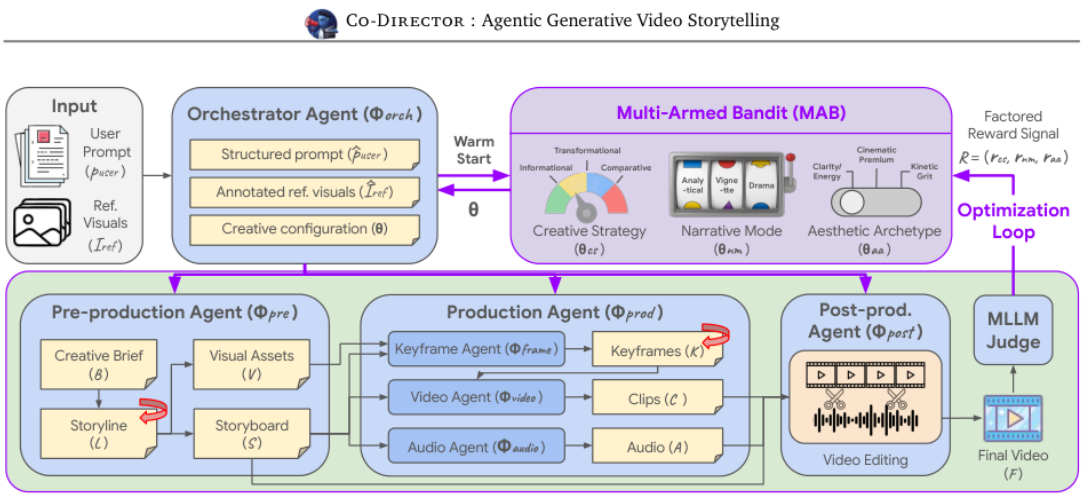

Figure 1 | **Co-Director** multi-agent pipeline: **(Top):** The Orchestrator Agent (Φ_{orch}) utilizes MAB to navigate a factored creative action space $\theta = (\theta_{cs}, \theta_{nm}, \theta_{aa})$. **(Bottom):** The Pre-Production Agent (Φ_{pre}) synthesizes a creative brief ($\mathcal{B}$), storyline ($\mathcal{L}$), and visual assets ($\mathcal{V}$) to establish a consistent storyboard ($\mathcal{S}$), which the Production Agent (Φ_{prod}) translates into synchronized keyframes ($\mathcal{K}$), video clips ($\mathcal{C}$), and audio ($\mathcal{A}$). An MLLM Judge evaluates the final video ($\mathcal{F}$) to generate a factored reward signal $R = (r_{cs}, r_{nm}, r_{aa})$, which is propagated back to the MAB to iteratively refine the creative configuration over T optimization loops.

Figure 1: Shows the complete Co-Director architecture with hierarchical multi-agent pipeline, MAB optimization of factored creative

space, and feedback loop structure that is central to the approach.

Limitations

- Relies on T=4 optimization iterations which may be insufficient for complex creative spaces requiring more exploration
- Uses fictional products in GenAd-Bench which may not fully capture real-world brand complexity and copyright constraints
- Local self-refinement limited to storyline and keyframes, not applied to video generation which could still produce temporal inconsistencies
- MAB formulation assumes creative parameters are independent, but artistic choices often have complex interdependencies